OPEN ENDED LAB REPORT

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DETERMINATION OF BEARING CAPACITY OF SOIL FOR DESIGNING THE FOOTING FOR A COLUMN.

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1-ABSTRACT:

This report presents a study on the determination of bearing capacity of soil for designing the footing for a column. The study involves conducting a direct shear test on the soil sample to determine the cohesion (c) and internal friction angle (ϕ) parameters. The bearing capacity of the soil is then calculated using the determined parameters and the Terzaghi's bearing capacity theory. The results of the study are presented, and the implications for designing the footing for the column are discussed.

2- INTRODUCTION:

The design of foundations for structures such as buildings, bridges, and towers requires a detailed understanding of the bearing capacity of the underlying soil. The bearing capacity of soil refers to the maximum load per unit area that the soil can sustain without experiencing failure. Soil failure can occur through mechanisms such as excessive settlement, shear failure, or a combination of both, potentially leading to structural instability. This study focuses on determining the bearing capacity of the soil to facilitate the design of an appropriate footing for a column, ensuring safety and stability while optimizing foundation dimensions.

SOIL SAMPLE:

We collected a soil as a sample in shelby tube for finding natural density of soil, And also collected 3kg soil for our test (Direct shear Test).

Location of soil sample:



3- LITERATURE REVIEW:

The determination of bearing capacity of soil is a fundamental aspect of geotechnical engineering, ensuring the stability and safety of structures such as buildings, bridges, and towers. Over the years, various theories and methodologies have been developed to estimate the bearing capacity of soil with increasing levels of precision and reliability.

One of the earliest and most widely used methods is **Terzaghi's bearing capacity theory (1)**, which provides the foundation for modern bearing capacity analysis. Terzaghi assumed that the soil behaves as a homogeneous, isotropic, and elastic material, and the footing is rigid, rough, and flat. His theory introduced key bearing capacity factors to calculate the ultimate bearing capacity of the soil, making it a simple and practical approach. Despite its assumptions, Terzaghi's method remains relevant in engineering practice due to its ease of application and robust framework **(1)**.

Expanding on Terzaghi's principles, **Meyerhof's bearing capacity theory (2)** incorporated additional parameters, including the shape, size, and depth of the footing. Meyerhof's model also considered the strength and density characteristics of the soil, providing a more comprehensive assessment of bearing capacity. This enhanced approach has been widely adopted for diverse foundation types and complex site conditions **(2)**.

The advent of laboratory-based testing methods has further refined bearing capacity analysis. Among these, the **direct shear test** has gained prominence as a reliable tool for determining the shear strength parameters of soil. According to ASTM D3080 **(5)**, the direct shear test measures soil properties such as cohesion ('c') and the internal friction angle (ϕ), which are integral to bearing capacity calculations. This test method is standardized and widely used to evaluate soil properties under consolidated drained conditions.

In recent years, researchers have emphasized integrating direct shear test results with classical theories. For instance, **Kumar and Sharma (2017)** demonstrated that cohesion and internal friction angle derived from direct shear tests could be used with Terzaghi's equations to estimate bearing capacity. Their findings underscored the importance of accurately determining soil strength parameters for reliable foundation design **(3)**.

Similarly, **Patel and Desai (2018)** investigated the application of direct shear test results in Meyerhof's bearing capacity framework. They highlighted that the test's

output could effectively determine the shear strength parameters, enhancing the precision of bearing capacity predictions. This research further validated the practical significance of combining laboratory tests with theoretical models to achieve accurate and efficient soil assessments **(4)**.

This study builds on these advancements by employing the direct shear test to determine the cohesion ('c') and internal friction angle (ϕ) parameters of soil samples. These parameters will then be utilized in conjunction with both Terzaghi's and Meyerhof's theories to estimate the soil's bearing capacity. By bridging experimental results and theoretical calculations, this approach aims to contribute to the existing body of knowledge and offer practical insights into soil bearing capacity determination.

4- METHODOLOGY:

Direct shear test:

A soil sample was collected from the field using Shelby tubes, and its density was determined by measuring the weight and volume of the sample. The weight of the soil sample was measured using a balance, and this value was used to calculate the density of the soil. The soil sample was then prepared for the direct shear test, which was performed according to the **ASTM D3080** standard. The direct shear test results were used to calculate the shear strength parameters of the soil, including the cohesion (c) and internal friction angle (ϕ), which were calculated using the equations $\mathbf{c = \tau / (\sigma * \tan(\phi))}$ and $\mathbf{\phi = \arctan(\tau / \sigma)}$, where τ is the shear stress, σ is the normal stress, c is the cohesion, and ϕ is the internal friction angle.

5- OBSERVATIONS AND CALCULATION:

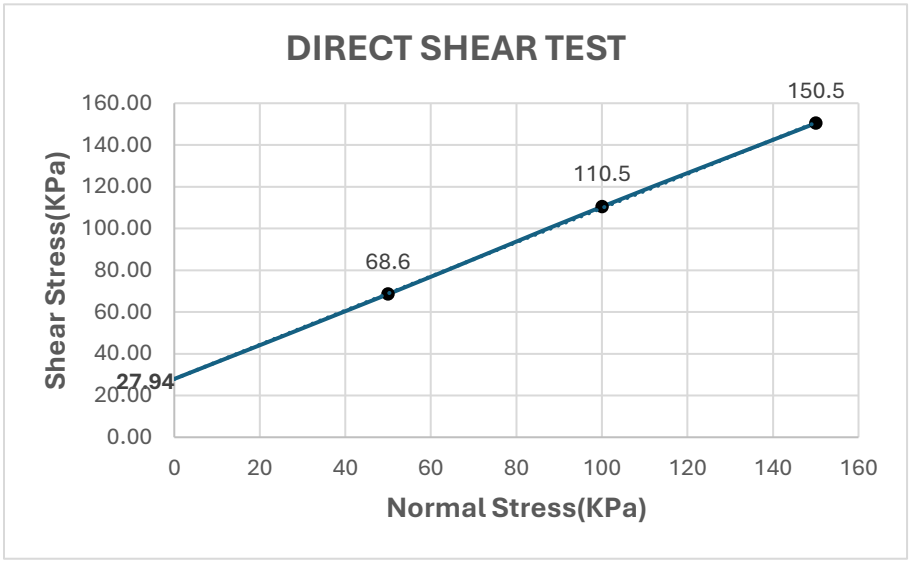
DIRECT SHEAR TEST CALCULATIONS:

A	Weight of specimen + Weight of ring	172 g
B	Weight of ring	40 g
C	Weight of specimen	132 g
D	Dia of specimen,D	6.18 cm
E	Area of specimen	30 cm^2
F	Initial height of specimen	2 cm
G	**Density of soil**	**2.2 g/cm^3**
H	Each Normal Load	1.275 kg
I	Proving Ring Calibration Factor	1.9048 KPa/div

Time(s)	H Dial(mm)	PR Dial(div)	PR Constant(KPa)	Shear Stress(KPa)
SAMPLE 01				
10	0.4	25	1.9048	47.62
20	0.8	34	1.9048	64.7632
30	1.2	36	1.9048	68.5728
40	1.6	36	1.9048	68.5728
SAMPLE 02				
10	0.4	32	1.9048	60.9536
20	0.8	45	1.9048	85.716
30	1.2	50	1.9048	95.24
40	1.6	58	1.9048	110.4784
50	2	58	1.9048	110.4784
SAMPLE 03				
10	0.4	34	1.9048	64.7632
20	0.8	53	1.9048	100.9544
30	1.2	69	1.9048	131.4312
40	1.6	79	1.9048	150.4792
50	2	79	1.9048	150.4792

Trial No	NL(kg)	NS(kg/cm²)	NS(KPa)	PR Dial(div)	Shear Stress(KPa)
1	1.275	0.51	50.031	36	68.5728
2	2.55	1.02	100.062	58	110.4784
3	3.825	1.53	150.093	79	150.4792

GRAPH:



RESULTS:

C=27.94 (kpa)
Φ=39.30

FOUNDATION DESIGN CALCULATION:

DATA:

Cohesion=c'= 27.94 KN/m^2
Internal friction=f= 39.30°
Factor of safety=F= 3
Depth of foundation=d= 1m
Unit weight of soil= 21.5 KN/m3
Load for designing the footing=**Q=500kip**=2226KN

CALCULATIONS:

According to Terzaghi's Table:

Nc = 85.97
Nq = 70.61
$N\gamma$ = 95.03

Terzaghi's Equation for Square Foundation:

$$qu = 1.3c'Nc+Nqq+0.4B\gamma N\gamma$$

qall = qu /F.O.S
Qgross = qall*B^2 OR qall = Qgross/B^2
Qgross/B^2= qu /FOS

2226/B^2 = 1/3 [(1.3c'Nc+Nqq+0.4Bg$N\gamma$)]
2226/B2 = 1/3 [(1.3*27.94*85.97) + (21.5*1*70.61) + (0.4*21.5*B*95.03)]

RESULTS:

By cross Multiplication & Hit and trial Method:

B = 3.6 ft

6-CONCLUSION:

The laboratory tests conducted to determine the geotechnical properties of the soil sample yielded values of cohesion (c) and internal friction angle (ϕ) as **27.94 kPa** and **39.30°**, respectively. Using these parameters and the bearing capacity equation, the base width of the square footing required to support a gross load of **500 kip** was calculated to be **3.6** feet. This result highlights the importance of accurate determination of soil properties in foundation design to ensure the stability and safety of structures.

7-RECOMMENDATIONS:

- Suitable for shallow foundations such as isolated or combined footings where moderate to high bearing capacities are required.
- Can support retaining walls due to its considerable internal friction angle, providing significant resistance to lateral earth pressures.
- Appropriate as a subgrade material for pavements due to its cohesive strength and stability under traffic loads.
- Useful in slopes and embankments due to its strength parameters, reducing the risk of shear failure under typical loading conditions.
- Applicable as structural fill in construction projects where stability and moderate bearing capacity are critical.
- Can be employed in temporary structures like braced excavations or trench supports due to its shear strength and resistance.

7-REFERENCES:

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